MIL-A-52685
11 May 1970

MILITARY SPECIFICATION

ADHESIVE: BONDING, CHLOROPRENE, AIR FIELD MEMBRANE SURFACING

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of adhesive for bonding chloroprene coated fabrics.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-P-704	- Pails, Metal: (Shipping, Steel, 1 through 12 Gallon).

Military

MIL-T-704	- Treatment and Painting of Materiel.

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of U. S. Military Property.
MIL-STD-147	- Palletized and Containerized Unit Loads, 40" x 48" 4-Way (Partial) Pallet Skids, Runners, or Pallet-Type Base.

FSC 5610

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(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

DEPARTMENT OF TRANSPORTATION (DOT)

DOT Federal Motor Vehicle Safety Standards.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

D-891	- Specific Gravity of Industrial Aromatic Hydrocarbons and Related Materials.
D-903	- Test for Peel or Stripping Strength of Adhesive Bonds.
D-1002	- Test for Strength Properties of Adhesives in Shear by Tension Loading. (Metal to Metal).
D-1084	- Test for Consistency of Adhesives.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Description. The adhesive shall be a rubber-based adhesive of noncuring polychloroprene type modified with a cyclic unsaturated hydrocarbon resin, usable without heating or additives.

3.2 First article.

3.2.1 Preproduction model. The supplier shall furnish a 5 gallon pail of adhesive for examination and tests within the time frame specified to prove, prior to starting production, that his production methods will produce adhesives that comply with the requirements of this specification. Examination and tests shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

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3.3 Materials. Material shall be as specified herein. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification.

3.4 Physical properties. The physical properties of the adhesive shall be in accordance with the requirements of table I when tested as prescribed in table II.

Table I. Physical Properties

Property	Minimum	Maximum
Total solids content (percent by weight)	20	30
Specific gravity	0.80	0.86
Viscosity (cps)	350	750
Ash content (percent by weight)	-	6
Tack range (minutes)	-	15 1/
Shear strength (psi)	-	-
Initial	200	-
Wet	200	-
Hot	200	-
Freeze thaw	200	
Joints utilizing 15 minutes open time	200	-
Peel strength (p/i)	-	-
Initial	7	-
Wet	6	-
Hot	6	-
Freeze thaw	6	-

1/ See note 7, table II.

3.5 Shelf life. The shelf life of the adhesive shall be not less than 1 year. The supplier shall furnish evidence acceptable to the contracting officer that the year old or older samples of the product he intends to supply will meet or exceed the requirements of this specification when inspected in accordance with 4.3. Adhesive shall not be more than 3 months old from date of manufacture to date of acceptance by the Government.

3.6 Identification marking. The adhesive container shall be identified in accordance with MIL-STD-130. The marking shall be stenciled on the container in accordance with MIL-T-704. The letters shall be Gothic capitals and the numerals shall be 0.75 inch high. In addition to the requirements of MIL-STD-130 figure 1, the date of manufacture, lot number, shelf life expiration date, weight and cubage shall be included.

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3.6.1 Warning marking. Each container shall be prepared in accordance with Code of Federal Regulations, DOT Federal Motor Vehicle Safety Standards, Title 49, Transportation, Part 173.

3.6.2 Instruction marking. Each container shall have a waterproof label securely attached to it with all necessary instructions for proper application of the adhesive.

3.7 Treatment and painting. Containers shall be treated and painted in accordance with MIL-T-704, type C. Color shall conform to color chip lusterless No. X34807.

3.8 Government-loaned property. The Government will loan the supplier adherend FSN 5680-921-6512 to be used in tests (see 4.5.2). This material is expendable.

3.8.1 Quantity of adherend. The adherend specified in 3.8 will be loaned on a piece basis whose size shall be approximately 1 foot by 3 feet. The number of pieces furnished shall be one piece for each 300 gallons or fraction thereof of adhesive manufactured for delivery. The minimum number of pieces shall be one piece for lots of less than 300 gallons of adhesive manufactured for delivery.

3.9 Workmanship. The adhesive shall be free of grit, lumps, skins, foreign matter and shall not gel or liver.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of preparation for delivery (see 4.6).

4.3 Preproduction inspection.

4.3.1 Examination. The adhesive shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The adhesive shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Lot. A lot shall be all adhesive manufactured as one batch and offered for delivery at one time.

4.4.2 Sampling.

4.4.2.1 Samples for examination. A sample for examination shall consist of a minimum of one 5 gallon pail of each lot marked for identification as specified in 3.6 and offered for delivery.

4.4.2.1.1 Examination sample size. For lot size less than 300 gallons, one 5 gallon pail will be selected. For lot size more than 300 gallons, one 5 gallon pail will be selected from each 300 gallons or fraction thereof.

4.4.2.2 Samples for test. A sample for test shall consist of a minimum of 4 quarts of adhesive selected from each lot manufactured for delivery.

4.4.2.2.1 Test sample size. If the lot is less than 300 gallons, one 4 quart sample will be selected for testing. If the lot is more than 300 gallons, one 4 quart sample will be selected from each 300 gallons or fraction thereof.

4.4.3 Examination. Sample selected in accordance with 4.4.2.1 shall be examined in accordance with 4.5.1. Presence of one or more defects shall be cause for rejection of the entire lot.

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4.4.4 Tests. Samples selected in accordance with 4.4.2.2 shall be tested as specified in 4.5.2. Failure of one or more tests shall be cause for rejection of the entire lot.

4.5 Inspection procedure.

4.5.1 Examination. The adhesive shall be examined for the following defects:

101. Workmanship not as specified.
102. Not marked in accordance with MIL-STD-130 and as specified in 3.6.
103. Evidence of shelf life submitted by supplier not acceptable.
104. Age of adhesive not as specified in 3.5.

4.5.2 Tests. The adhesive shall be tested in accordance with the methods prescribed in table II. Nonconformance with requirements of table I shall constitute failure of the test.

4.5.2.1 Adherend. The adherend used will be Government-loaned (see 3.8).

4.5.2.2 Fabrication of specimen joints. Joints shall be fabricated by the application of the adhesive in a single coat of 0.008 to 0.012 inch in wet thickness to each adherend. Join the adherends when the adhesive has become tacky and allow to stand for 15 minutes before rolling lightly to remove air pockets and excess adhesive. Condition and test as indicated in applicable test method.

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Table II. Physical Properties Tests and Methods

Property	Test Methods
Total solids content	As specified 8/
Specific gravity	ASTM D-891, method B (77 ± 5 degrees).
Viscosity	ASTM D-1084, method B.
Ash content	As specified 1/
Shear strength 2/& 3/	ASTM D-1002
Initial	ASTM D-1002
Wet 4/	ASTM D-1002
Hot 5/	ASTM D-1002
Freeze thaw 6/	ASTM D-1002
After tackiness test 7/	ASTM D-1002
Peel strength 2/	ASTM D-903
Initial	ASTM D-903
Wet 4/	ASTM D-903
Hot 5/	ASTM D-903
Freeze thaw 6/	ASTM D-903

1/ Place approximately 5 grams of adhesive as supplied in preweighed porcelain crucibles. Weigh crucible plus adhesive. Ash in 1070° F. plus or minus 5° F. furnace for 12 hours.

$$\text{Weight percent ash} = \frac{\text{weight of residue}}{\text{weight of adhesive}} \times 100$$

2/ A minimum of five specimens representing five different glue lines shall be tested. The reported value will be the arithmetic average of the five specimens tested.

3/ All shear strength specimens shall be initially conditioned as specified in ASTM D-903. The adherend used shall be as prescribed in 4.5.2.1 in lieu of metal.

4/ Immerse in distilled water for 48 hours at 75° F. plus or minus 2° F. Remove, blot dry and test within 15 minutes.

5/ Condition for 48 hours at 160° F. plus or minus 5° F. Cool for no more than 2 hours at 75° F. plus or minus 5° F. and test.

6/ Condition for 4 hours at minus 70° F. plus or minus 5° F., allow to thaw for 3 hours at 75° F. plus or minus 5° F. and test.

7/ Apply the adhesive in accordance with the suppliers instructions on adherends (see 4.5.2.1). Adhesive must become tacky within 1 minute and remain tacky for 14 additional minutes. Shear strength will be determined for five specimens joined after remaining open for 15 minutes.

8/ Pour approximately 10 grams of the sample into a low form weighing bottle, cover and weigh. After removing the cover, apply heat at 160° F. plus or minus 2° F. until the sample reaches a constant weight. Then cool the sample in a desiccator and weigh.

Calculate the percentage of solids as follows:

$$\text{TOTAL Solids Percent} = \frac{\text{WT of Residue}}{\text{WT of Sample}} \times 100$$

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4.6 Inspection of preparation for delivery.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a filled pail prepared for shipment, shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105. AQL shall be 2.5 percent defective.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects:

105.	Materials and containers not as specified.	Each incorrect material or container shall be considered one defect.
106.	Marking illegible, incorrect or incomplete.	
107.	Painting not as specified.	

5. PREPARATION FOR DELIVERY

5.1 Packaging (see 6.5). The adhesive shall be packaged in 5-gallon tight head pails conforming to PPP-P-704, type I, class 4, equipped with a push-pull spout. Closure shall be effected by machine or by ratchet type handtool designed for tightening the screwcap fitting to insure the closure is tightened to the maximum practical extent. The screwcap shall then be secured with a closure seal as specified in PPP-P-704.

5.2 Packing (see 6.5). The pails specified in 5.1 shall not require packing. When specified, 5-gallon pails of adhesive shall be palletized in accordance with MIL-STD-147, load type III, figure 5a and 5b.

5.3 Marking. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The adhesive covered by this specification is intended to be used for the joining, by bonding, of airfield surfacing membranes.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Time frame required for submission of preproduction model (see 3.2.1).
(c) When pails shall be palletized (see 5.2).

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6.3 Preproduction model. Any changes or deviations of production adhesive from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the supplier of his obligation to furnish adhesive conforming to this specification.

6.4 Government-loaned property. The contracting officer should arrange to loan the property specified in 3.8.

6.5 Only one level of packaging and packing is required. The 5-gallon container is the sole acceptable container based on end use requirements. For purposes of packaging/packing level, marking A/A may be used.

Custodians:

Army - ME
Navy - YD
Air Force - 84

Preparing activity:

Army - ME

Project No. 5680-0034

Review activities:

Army - MR, CE

User activity:

Navy - MC

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-A-52685 Adhesive: Bonding, Chloroprene, Air Field Membrane Surfacing

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

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